CHAPTER

11

Low-Level DNA Testing: Issues, Concerns, and Solutions

INTRODUCTION TO ISSUES AND HISTORICAL OVERVIEW

The polymerase chain reaction (PCR) is very sensitive and STR typing results have been demonstrated from as little DNA as that contained in a single collected cell (Findlay et al. 1997). This capability has encouraged attempts to try to recover DNA profiles from touch evidence that might be helpful in a case. However, this low-level DNA analysis can sometimes push the envelope of what constitutes reliable results unless measures are taken to demonstrate reproducibility of allele calls. This chapter will address approaches to improve DNA sensitivity and efforts made to strengthen reliability with low-level DNA testing.

Terms and Definitions

First, it is worth discussing a few terms that are commonly used to describe low-level DNA testing. The most common is probably *low copy number*—often shortened to the abbreviation LCN. More recently, some labs have begun referring to LCN as *low template DNA* (LT-DNA or LTDNA) to reflect the fact that very small amounts of DNA are being examined. The term LCN typically relates to using a higher sensitivity techique while the term LTDNA is meant to focus on the sample. Others have called this technique the analysis of DNA of low quality and quantity (DNALQQ). Throughout this chapter this topic is primarily called "low-level DNA" but these other terms are interchangeable in most instances.

Although there is not really a rigid line dividing low-level DNA from conventional STR typing (Gill & Buckleton 2010), a number of different definitions have been debated. The definitions can be divided into several groups: (1) those that refer to the amount of DNA tested in the PCR reaction (e.g., $<$100 pg or $<$200 pg) based on some form of quantitation assay, (2) a methodology usually involving increasing the number of PCR cycles (e.g., 34 cycles instead of 28 cycles) to improve assay sensitivity, and (3) DNA profile appearance that exhibits an increased imbalance of observed alleles (e.g., heterozygote peak height ratios below 60%). In all these definitional approaches, it is recognized that data reliability is inferior

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when lower amounts of DNA are tested and thus additional measures must be taken to improve the chance of obtaining results that accurately reflect the sample being examined.

LTDNA has been defined as "an ultra sensitive technique that has the potential to yield a DNA profile from sub-optimal biological samples" (Caddy et al. 2008). Another definition prepared by a U.S.-based committee in January 2009 is more explicit: "Based upon a laboratory's internal validation, any DNA typing results generated from limited quantity and/or quality DNA template using conditions that have demonstrated increased stochastic effects are defined as LCN or Low Level DNA analyses. The stochastic effects which may be observed in DNA samples subjected to these conditions include allelic drop-in or drop-out, increased stutter and increased intra-locus peak height imbalance. When used to analyze limited quantity and/or quality DNA template below the stochastic thresholds, LCN conditions may include the following: additional amplification cycles, post-amplification purification, reduced reaction volume, injection enhancement by increased voltage or time, and nested PCR" (SWGDAM Ad Hoc LCN Committee 2009, personal notes). Here the focus is more on performance issues rather than specifying a particular amount of DNA under which less reliable results may be obtained.

Many laboratories have demonstrated the ability to obtain DNA profiles from very small amounts of sample. Low-level DNA testing typically refers to examination of less than 100 pg of input DNA, or about 15 diploid cells (Gill et al. 2000). Others have raised their definitional LCN threshold to 200 pg or about 30 diploid cells (Caddy et al. 2008, Budowle et al. 2009).

Uses with Touch DNA and Potential Transfer Issues

The capability of obtaining a useful DNA profile is often only limited by the ability of the forensic investigator to find and collect the appropriate evidence. Remarkably, DNA profiles may be obtained from fingerprint residues due to cells that are left on the objects that are touched (van Oorschot & Jones 1997). DNA technology may permit the handles of tools used in crimes, such as knives or guns, to be effectively evaluated and used to link a perpetrator to his crime. A December 2010 review article on forensic trace DNA cites over 200 articles on this topic (van Oorschot et al. 2010).

It is important to realize that when trying to work with extremely low levels of DNA template, recovered DNA profiles may not be associated with the crime event itself but rather have been left innocently before the crime occurred (Gill 2001). Secondary transfer of skin cells due to casual contact such as hand shaking has been demonstrated to occur in controlled laboratory settings (Farmen et al. 2008). This phenomenon occurs to a variable degree depending on what kind of a "shedder" the individuals are (Lowe et al. 2002). A more recent study found no good shedders out of 60 volunteers tested (Phipps & Petricevic 2007). As noted in Chapter 1, potential transfer of DNA is highly variable depending on surface texture and sample moisture.

LT-DNA Labs and Published Studies

While many laboratories may be performing testing with low amounts of DNA, currently only a few laboratories are formally conducting LT-DNA casework with specific enhanced detection protocols. These LT-DNA laboratories include the Forensic Science

Service (Birmingham, UK), LGC Forensics (Middlesex, UK), Orchid Cellmark (Abingdon, UK), the Netherlands Forensic Institute (NFI, The Hague, The Netherlands), Institute for Environmental Science & Research (ESR, Auckland, New Zealand), the Swedish National Laboratory of Forensic Science (Linköping, Sweden), the German Bundeskriminalamt (BKA, Wiesbaden, Germany), the International Commission on Missing Persons DNA Laboratory (ICMP, Sarajevo, Bosnia and Herzegovina), the Armed Forces DNA Identification Laboratory (Rockville, Maryland), the University of North Texas Center for Human Identification (Ft. Worth, Texas), and the New York City Office of Chief Medical Examiner (NYC OCME) Forensic Biology Laboratory. The applications of LT-DNA testing range from missing persons investigations to forensic casework.

In addition to the early work by the FSS (Gill et al. 2000, Whitaker et al. 2001), validation studies on LT-DNA techniques have been published by scientists from NFI (Kloosterman et al. 2003, Western et al. 2009, Benschop et al. 2010), LGC (Forster et al. 2008, Gross et al. 2009), Orchid Cellmark (Roeder et al. 2009), NYC OCME (Caragine et al. 2009) and ESR (Petricevic et al. 2010). John Buckleton speculates that more LT-DNA validation articles have not been published because journals may not consider the techniques sufficiently novel (Buckleton 2009).

Historical Overview of Low-Level DNA Testing

Since the late 1990s, there have been numerous efforts with low-level DNA testing (Table 11.1). Single-cell analysis for clinical applications were being performed in 1995 (Findlay et al. 1995) and demonstrated with STR typing by 1997 (Findlay et al. 1997). Pierre Taberlet and colleagues at the French Laboratoire de Biologie des Populations d'Altitude first published the concept of performing replicate PCR amplifications to improve reliability with low quantities of DNA and only recording an allele if it was observed at least twice (Taberlet et al. 1996). As will be seen later in the chapter, this replicate amplification strategy has become the core feature of reliable low-level DNA testing.

Beginning in 1999, the FSS led the way with application of LCN to forensic DNA testing (Gill et al. 2000, Whitaker et al. 2001). The FSS also studied secondary transfer (Lowe et al. 2002) and eventually developed software to probabilistically model stochastic effects (Gill et al. 2007). In December 2007, the Omagh bombing case outcome (D.N.A. Box 11.1) led to a temporary suspension of LCN testing in the UK and the formation of a Forensic Regulator to oversee quality assurance efforts.

Within the U.S., the FBI urged caution and discouraged efforts towards LCN DNA analysis for criminal casework (Budowle et al. 2001). However, the NYC OCME began research with low-level DNA testing in 2001 and casework in 2006 after approval from the New York Forensic Science Commission. Over the past decade, they have published improvements to DNA extraction (Schiffner et al. 2005), procedures for decontaminating plasticware and water used in PCR reactions (Tamariz et al. 2006), and a description of their extensive validation studies (Caragine et al. 2009).

LCN Critics and the Omagh Trial

As with any technique that is subjected to scrutiny in court proceedings, there are often individuals with opposing points of view testifying for the defense. Critics of low-level DNA

TABLE 11.1 Timeline for Low-Level DNA Work.

Year	Events
1995	Single cell analysis for clinical applications (Findlay et al.)
1996	Reliable results from pg quantities using replicate amplifications (Taberlet et al.)
1997	DNA results from single cells (Findlay et al.); DNA results from touched objects (van Oorschot & Jones)
1998	Nested PCR approach (Strom & Rechitsky)
1999	SGM Plus kit released; FSS begins LCN casework with 34-cycle SGM Plus
2000	FSS LCN paper published (Gill et al.)
2001	FSS SGM Plus LCN paper published (Whitaker et al.); NYC OCME begins research; FBI urges caution (Budowle et al.)
2002	Secondary transfer studied (Lowe et al.); LCN data accepted on UK NDNAD
2003	Dutch publications on advantages & limitations of LCN (Kloosterman & Kersbergen)
2004	Whole genome amplification ineffective with low-level DNA (Schneider et al.)
2005	Modeling experiments (Gill et al.); NYC OCME receives approval from New York Forensic Science Commission for LT-DNA testing
2006	NYC OCME goes online with LT-DNA testing; ISFG DNA Commission publication on mixtures (recommendation #9 discusses LT-DNA)
2007	LoComationN software described (Gill et al.); Judgment in the Omagh bombing case (*R v Hoey*); interim suspension of FSS LCN
2008	Caddy report issued; Forensic Regulator response; LGC publication (Forster et al.)
2009	Critical review (Budowle et al.) and OCME validation (Caragine et al.) published; ISFG and ISHI conference debates; UK appeal (*R v Reed & Anor*) ruling in favor of FSS LCN
2010	SWGDAM committee established; NYC *Frye* ruling (*NY v Megnath*) allows OCME LT-DNA results; Wallace case in New Zealand finds LCN not inherently unreliable
2011	*Forensic Science International: Genetics* publishes several Letters to the Editor with an accompanying editorial requesting more science and less quarrelling on the topic

testing have included Professor Allan Jamieson of the Forensic Institute in Glascow, Scotland (http://www.theforensicinstitute.com/), Professor Dan Krane of Wright State University in Dayton, Ohio and founder of Forensic Bioinformatics (http://www.bioforensics.com/), and Professor Bruce Budowle of the University of North Texas Institute of Investigative Genetics in Fort Worth, Texas, who for many years was a senior scientist at the FBI Laboratory. At many of the trials involving LT-DNA, one or more of these three critics have voiced their concerns regarding low-level DNA analysis.

By early 2010, the UK's Forensic Science Service, which pioneered the approach more than a decade ago (Gill et al. 2000), reportedly had used LCN analysis on over 21,000 criminal cases (Gilbert 2010). However, the FSS efforts with LCN are most often connected to the 2007 Omagh bombing trial (D.N.A. Box 11.1) and the subsequent fallout that lead to

D.N.A. BOX 11.1

THE OMAGH BOMBING TRIAL

On the afternoon of Saturday, August 15, 1998, a car bomb exploded in the middle of a busy shopping center in the Northern Ireland town of Omagh. The powerful explosion destroyed the downtown area, killed 29 people, and injured over 200 others. It is considered by the media as Northern Ireland's worst terrorist attack. Pictures and information regarding those killed at Omagh are available at http://www.wesleyjohnston.com/users/ireland/past/omagh/dead.html.

Within a few weeks, the police arrested 12 men in connection with the crime, but all were subsequently released without charges due to lack of evidence against them. In February 1999, the police arrested Colm Murphy, a local building contractor, for questioning in connection with the Omagh bombing. Murphy was later charged and subsequently convicted in 2002 to 14 years in prison for conspiring to cause the Omagh bombing. However, in 2005, his conviction was overturned on appeal and in February 2010, Murphy was acquitted in a retrial in spite of being found liable for the bombing along with three others in a 2009 civil court ruling.

In September 2006, Murphy's nephew—an electrician named Sean Hoey—was placed on trial with 29 counts of murder as well as terrorism and explosive charges. A key part of the trial involved low-level DNA analysis by the Forensic Science Service on pieces of tape from the bomb timer and explosive wiring. When the controversial trial concluded on December 20, 2007, Mr. Hoey was found not guilty of all 58 charges brought against him.

But perhaps even more significantly the judge—Justice Weir—lashed out against the handling of the DNA evidence. Court records state: "The evidence establishes that the arrangements within the police in 1998 and 1999 for the recording and storage of items were thoroughly disorganized" (*R v Hoey*, paragraph 51). As noted in Chapter 1 of this book, if the evidence handling and chain-of-custody (bagging, labeling, packaging, storage, tracking, and transmission of evidence) is in doubt, then a case may be in jeopardy regardless of the DNA test results. In addition, because evidence in this case was handled prior to the development and implementation of the FSS LCN procedures, there were no programs in place by the crime scene investigators for special handling of evidentiary items that would later be scrutinized by the FSS with their highly sensitive 34-cycle LCN technique.

Justice Weir also expressed "concern about the present state of the validation of the science and methodology associated with LCN DNA and, in consequence, its reliability as an evidential tool." In particular, his ruling stated, "I am not satisfied that the publishing of two journal articles [(Gill et al. 2000, Whitaker et al. 2001)] describing a process invented by the authors can be regarded without more as having 'validated' that process for the purpose of its being confidently used for evidential purposes."

Media around the world picked up on the purported problems with LCN and suddenly the *The Queen v Sean Hoey* (*R v Hoey*) case became synonymous with poor DNA test results.

Sources:

R v Hoey (2007). Available at http://www.denverda.org/DNA_Documents/Hoey.pdf.
http://en.wikipedia.org/wiki/Omagh_bombing
http://en.wikipedia.org/wiki/Colm_Murphy
http://news.bbc.co.uk/2/hi/uk_news/northern_ireland/7154221.stm
http://news.bbc.co.uk/2/hi/uk_news/northern_ireland/7149505.stm

a review of the FSS efforts, the "Caddy Report" recommendations (D.N.A. Box 11.2), and establishment of a Forensic Science Regulator in the United Kingdom.

The Caddy Report and the Forensic Science Regulator

The Omagh bombing case (*R v Hoey*) judgment against LCN DNA testing in December 2007 sent shockwaves through forensic service providers in the United Kingdom, particularly the Forensic Science Service. An interim suspension was put on all LCN DNA analysis in criminal investigations in England and Wales while an internal review of on-going cases involving the FSS and LCN DNA was performed between December 21, 2007 and January 14, 2008.

The Crown Prosecution Service (CPS) that conducted this internal review concluded "that LCN DNA analysis provided by the FSS should remain available as potentially admissible evidence" (CPS Press Release, January 14, 2008). The CPS press release went on to state that "at present, there is no reason to believe that there is any inherent unreliability in the LCN DNA analysis process provided that it is carried out according to the prescribed processes, and that the results are properly interpreted. In its work so far, the review has found nothing that would indicate any serious flaws in the scientific principles."

A Forensic Science Regulator was established to help strengthen quality assurance processes in the UK. In addition, an expert review of LT-DNA was conducted from November 2007 to April 2008. Because this review was led by Professor Brian Caddy of Strathclyde University (Glasgow, Scotland), it is typically referred to as the "Caddy Report". Also serving with Professor Caddy were Dr. Graham Taylor of Cancer Research UK (Leeds, England) and Dr. Adrian Linacre, who was from Strathclyde University at the time.

The authors of the Caddy Report state that "it is our opinion that LCN and LTD[NA] are extensions of the internationally accepted process of standard DNA profiling" (Caddy et al. 2008, section 7.2). They further conclude that "reservations [with regard to the LCN technique] have been allayed from a study of the raw data produced by the FSS, recent experimental work conducted by the FSS and also from detailed information submitted by the other [UK] forensic science providers which clearly demonstrate the soundness of LTDNA analysis (including LCN) *providing all the appropriate conditions are met*" (emphasis added).

The Caddy Report emphasizes that "any LTDNA profile should always be reported to the jury with the caveats: that the nature of the original starting material is unknown; that the time at which the DNA was transferred cannot be inferred; and that the opportunity for secondary transfer is increased in comparison to standard DNA profiling" (Caddy et al. 2008, section 7.4). The report concluded with 21 recommendations (D.N.A. Box 11.2).

In May 2008, about a month after the Caddy Report was issued, the Forensic Science Regulator Andrew Rennison responded. Mr. Rennison concluded that after having considered the Caddy Report: "I am content that the science underpinning the LTDNA analytical services, as provided to the [criminal justice system], is sound and that the three forensic science suppliers [FSS, LGC Forensics, Orchid Cellmark Ltd] offering such services have properly validated their processes. There is no flaw inherent in the process which prevents its use with the [criminal justice system]" (Forensic Science Regulator 2008).

D.N.A. BOX 11.2

CADDY REPORT RECOMMENDATIONS

The Caddy Report, named for Professor Brian Caddy who led the investigation, was an expert panel review of low template DNA (LTDNA) work conducted from November 2007 to April 2008 as a follow-up to the concerns raised duing the Omagh bombing trial (D.N.A. Box 11.1). The Caddy report concluded with 21 recommendations provided primarily to the UK Forensic Science Regulator, which are summarized below. These 21 recommendations can be grouped into broad areas of training, quality standards, and research.

1. Institute a national education program teaching the advantages and limitations of LTDNA for crime scene investigators
2. Come to an agreement on what constitutes LTDNA success
3. Institute appropriate training programs and set standards to enable police forces to have a full grasp of what constitutes LTDNA analysis, and how such samples are to be collected and stored especially in relation to issues of contamination and the likelihood of success
4. Monitor the use of DNA quantification procedures
5. Ensure collection of DNA profiles from all serving operational police officers and crime scene personnel as a means of eliminating irrelevant DNA profiles
6. Establish a national standard for "DNA clean" consumables, especially in relation to crime scene recovery kits; only kits that meet such a standard should be used by police forces
7. Ensure the batch testing of all DNA reagents to verify that they are DNA free prior to their use
8. Oversee compliance with standards of competence for LTDNA laboratory specialists and suggest modifications where needed to training programs and record keeping
9. Police forensic science laboratories need to be accredited and comply with ISO 17025
10. Quantification of the material extracted for analysis from LTDNA samples must be undertaken
11. Develop a consensus from all the forensic science providers on how profiles and mixed profiles are to be interpreted—and then monitor their implementation and encourage openness in the availability of information that may have an impact on the way DNA profiles are interpreted in the context of a case; there needs to be a national agreement on how LTDNA profiles are to be interpreted especially in relation to "allele drop in and out," stochastic effects, inhibition, and mixtures
12. Institute a regular program of inspections of documentation associated with all validations
13. State appropriate caveats in witness statements/court reports when LTDNA analyses have been undertaken
14. Gather an independent internationally recognized expert panel to review validation data on a regular basis; copies of raw validation data should be lodged with the Regulator before the method is introduced into service
15. Assess the advantages and disadvantages of different approaches

D.N.A. BOX 11.2 (*cont'd*)

to LTDNA analysis; mention of a forthcoming paper that is now published (Forster et al. 2008)

16. Improve existing guidelines and standards with an advisory group; develop documentation that would guide the courts in the interpretation of evidence
17. Consider alternative technologies, such as the use of next generation sequencers; tap into the expertise of the Wellcome Trust Sanger Institute and look into mechanisms for improving funding
18. Seek funding for independent research and validation that is open to national competition
19. Establish national minimum technical standards for extraction, quantification/dilution and interpretation criteria that will be agreed upon by all forensic science providers
20. Quantify the problem of financing forensic science and explore mechanisms to correct any problems the inquiries reveal
21. Explore the means of establishing a professional forensic science provider's organization in order to develop mutually agreed standards

Source:
Caddy, B. et al. (2008) Review of the science of low template DNA analysis. Available at http://police.homeoffice.gov.uk/publications/operational-policing/Review_of_Low_Template_DNA_1.pdf.

Scientific Conference Debates in 2009 & 2010

The debate of how to handle LT-DNA samples reached a boiling point at two large international scientific conferences in the fall of 2009. The International Society for Forensic Genetics (ISFG) concluded its September 15–18, 2009 conference in Buenos Aires, Argentina with presentations by Adrian Linacre, Bruce Budowle, and Peter Gill. Short articles describing the presenter's positions have been published in volume 2 of the *Forensic Science International: Genetics Supplement Series* (*Progress in Forensic Genetics 13*) available at http://www.fsigeneticssup.com/.

A month later the International Symposium on Human Identification (ISHI) sponsored by Promega Corporation held an LCN panel on October 15, 2009 in Las Vegas, Nevada. This LCN panel consisted of Charlotte Word, Bruce Budowle, Theresa Caragine, Angela van Daal, Gillian Tully, Brad Leventhal, and John Butler. Information on most of the presenter's positions is available in the March 2010 and September 2010 issues of Promega's *Profiles in DNA* available at http://www.promega.com/profiles/. Some of the presentations are also available on the NIST STRBase website at http://www.cstl.nist.gov/biotech/strbase/LTDNA.htm.

In September 2010, a follow-up debate occurred at the Australian-New Zealand Forensic Science Society (ANZFSS) meeting in Sydney, Australia with Peter Gill and John Buckleton supporting properly performed LT-DNA work and Bruce Budowle and Angela van Daal sharing their concerns regarding LCN in general. Adding to the drama of this debate was a series of Letters to the Editor that appeared throughout 2010 on the http://www.fsigenetics.com/inpress website.

These Letters to the Editor were published in the January 2011 issue of *Forensic Science International: Genetics* along with an editorial effectively calling for a "cease fire" and encouraging more scientific data to improve the field rather than revisiting the same arguments over and over (Schneider et al. 2011).

Admissibility within UK, U.S., and New Zealand Courts

In December 2009, the England and Wales Court of Appeals Crown Court ruled that LCN testing was reliable in an important UK appeals case in spite of concerns raised by Allan Jamieson and Bruce Budowle (*R v Reed & Anor 2009*). Within the United States, an important *Frye* hearing in February 2010 ruled in favor of admissibility of LCN DNA typing. This New York Supreme Court case, *The People of the State of New York v. Hemant Megnath*, involved extensive testimony by scientists from the NYC OCME, the FSS, and the DNA subcommittee chair of the New York Forensic Science Commission as well as defense expert testimony from Dan Krane and Bruce Budowle. The court found "that LCN DNA profiling as conducted by the OCME is *not a novel* scientific technique" and "that LCN DNA testing as conducted by the OCME is generally accepted as reliable in the forensic scientific community" (NYC LCN *Frye* hearing, Feb 2010; emphasis in the original).

As of early 2011 there have not been a lot of rulings for or against low template DNA testing because the technique has so far been used extensively in only a relatively few jurisdictions. Most of the court decisions have focused on the big-picture view of the method. For example, a March 2010 New Zealand appeals court ruling, *Michael Scott Wallace v. The Queen*, found "that LCN DNA evidence is not inherently unreliable" (Wallace v. Queen 2010). Although court judgments have ruled in favor of the admissibility and the capability of low-level DNA testing, the specifics of a laboratory's protocol or performance in an individual case may come under scrutiny. Thus, scientists need to be able to demonstrate that their methods have been validated so that the limits of the technique are known, understood, and applied.

STRATEGIES FOR IMPROVING SENSITIVITY

Improved sensitivity in a detection technique is usually a valuable asset to enable results to be obtained from limited biological evidence. Not satisfied with failure to obtain a result with low amounts of DNA template, some laboratories have decided to push the envelope and apply what could be termed "enhanced interrogation techniques" (Butler & Hill 2010). In fact, DNA testing has been successfully applied down to the single-cell level (D.N.A. Box 11.3).

The advantages and disadvantages of various strategies for improving sensitivity with low-level DNA samples are shown in Table 11.2. Each of these strategies will be briefly discussed in the following section.

Increasing detection sensitivity is like turning up the volume on a radio. This action results in improved ability to hear sound at a greater distance. However, increasing the volume may distort the quality of the sound as it becomes louder. Contamination of low amounts of DNA from exogenous sources and the possibility of picking up cells from secondary transfer (see Chapter 1) represent noise that can interfere with detecting the true signal when sensitivity of PCR is heightened.

D.N.A. BOX 11.3

SINGLE CELL STR TYPING SUCCESS RATES OBSERVED IN A 1997 STUDY

The first demonstration of single-cell analysis with short tandem repeat (STR) loci is instructive (Findlay et al. 1997). This work involved collaboration between the Department of Molecular Oncology at the University of Leeds and the Forensic Science Service (FSS), both in the UK. A total of 226 buccal cells from four different individuals were isolated using micromanipulation techniques. These individual cells were then amplified with the FSS Second Generation Multiplex (SGM) assay using AmpliTaq Gold and 34 cycles to improve sensitivity. The SGM assay amplified six STR markers (FGA, TH01, vWA, D8S1179, D18S51, and D21S11) as well as amelogenin for sex-typing purposes (see Chapter 5). Single-cell results were compared with known DNA profiles from the four donors. While a full (and correct) DNA profile was observed in 50% of the single-cell tests, some kind of result was obtained in 91% of the tests. Some of these single-test results did not accurately represent the original donor DNA profile due to allele drop-out or drop-in.

A summary of the Findlay et al. (1997) results are shown below:

Number of single cells analyzed	226
Results obtained	206 (91%)
Full STR profile (amelogenin, 6 STRs)	114 (50%)
Acceptable profile (amelogenin, 5 to 6 STRs)	144 (64%)
Partial profile (1 to 4 STRs)	62 (27%)
Amplification failure	20 (9%)
Allele drop-out	88 (39%)
Surplus alleles (apparent mixture)	28 (12%)
False alleles (allele drop-in)	11 (5%)

The enhanced sensitivity that brought 91% amplification success also resulted in allele drop-out and drop-in due to stochastic effects (see Chapter 4). Allele drop-in (noted as "surplus alleles" and "false alleles" in the original article) occurred in approximately 17% of the PCR amplifications. With the "surplus alleles" situation, additional alleles were present in combination with the true donor alleles creating an apparent mixture. In the "false alleles" situation, the extra alleles replaced the true alleles, which had dropped out. This study found that with the 39% allele drop-out observed, each allele had about a 10% probability of being missing. Thus, wild-card designations permitting any allele to be considered in statistical calculations are important to properly assess evidence analyzed from low-level DNA. This work laid the groundwork for the future FSS developments in low-copy-number DNA analysis.

Source:

Findlay, I., et al. (1997). DNA fingerprinting from single cells. Nature, *389, 555–556.*

TABLE 11.2 Advantages and Limitations of Different Sensitivity Improvement Strategies. Most Strategies Involve Either Creating More PCR Product during DNA Amplification or Injecting More PCR Product into the Capillary Electrophoresis (CE) to Improve Detection.

Strategy	Advantage	Limitation	Example Reference
Increased number of PCR cycles	Creates more PCR product	Allele drop-in possible	Gill et al. (2000)
Post-PCR sample desalting	Improves injection of PCR product into CE capillary	Extra expense to sample processing; stochastic threshold needs to be raised to avoid false homozygote designations	Smith & Ballantyne (2007)
Increased CE injection	Improves amount of PCR product injected into CE capillary	Stochastic threshold needs to be raised to avoid false homozygote designations	Westen et al. (2009)
Reduced volume PCR	Concentrates PCR product relative to amount subjected to CE analysis	PCR inhibitors may be concentrated causing amplification failure; pipetting precision can be more challenging	Leclair et al. (2003)
Nested PCR	Creates more PCR product	Prone to contamination because tubes are opened to add second round of primers and reagents	Strom & Rechitsky (1998)
PCR enhancements (primer positions, polymerase concentration, etc.)	Creates more PCR product	Stochastic threshold needs to be raised to avoid false homozygote designations	MiniFiler kit (see Chapter 10)
mtDNA	Higher starting copy number per cell	Lower power of discrimination; cannot resolve individuals from same maternal lineage	Holland & Parsons (1999)

Increased Number of PCR Cycles

In the late 1990s, the United Kingdom's Forensic Science Service pioneered the application of LCN analysis through increasing the number of PCR cycles in order to improve DNA detection sensitivity (Gill et al. 2000, Whitaker et al. 2001). Instead of using their STR kit manufacturer's recommended 28 cycles, which has a theoretical yield of 67 million copies for each target DNA sequence, an additional six cycles (34 total) are run to provide a theoretical yield of 4.3 billion copies or a 64-fold improvement in sensitivity (see Table 4.1). A more recent approach to high-sensitivity DNA testing uses a three-cycle signal enhancement to provide a theoretical 16-fold improvement in sensitivity (Caragine et al. 2009).

This increase in PCR amplification cycles enables STR typing to routinely obtain results with samples containing less than 100 pg of DNA template (Figure 11.1). However, application of low level DNA results should be approached with caution due to the possibilities of

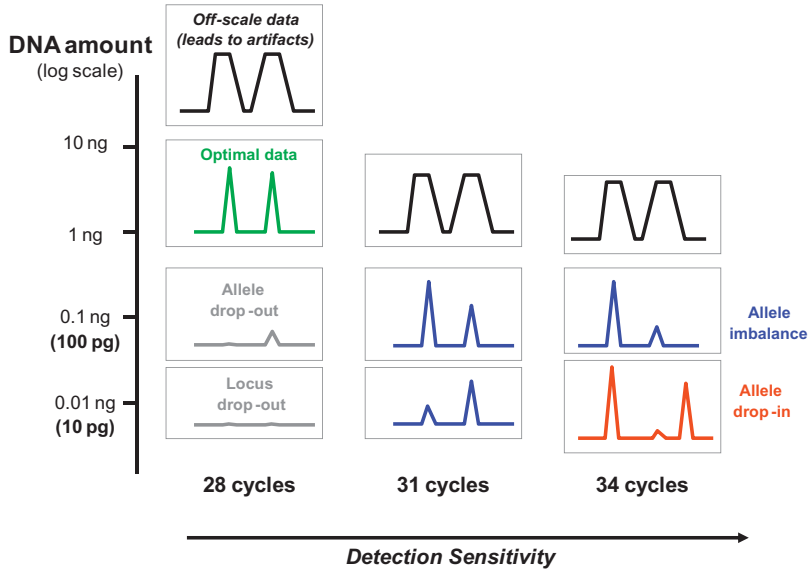


FIGURE 11.1 Illustration of hypothetical results at a heterozygous locus (two alleles expected in each box) with various levels of input DNA and detection sensitivity that is modulated through increasing the number of PCR cycles. Note that off-scale data occurs with lower amounts of DNA as the detection sensitivity is increased.

allele drop-out, allele drop-in, and increased risks of collection-based and laboratory-based contamination.

The original FSS approach was to run 34 PCR cycles with the SGM Plus kit without performing prior DNA quantitation. Thus, some "LCN" samples could be a nanogram or more in amount and generate off-scale data (Figure 11.1). The NFI developed a 28-cycle protocol with an extra six-cycle option (28 + 6) after the addition of fresh DNA polymerase (Kloosterman & Kersbergen 2003). The 28 + 6 approach permits screening samples at the 28-cycle level to prevent high amounts of DNA from being tested with the full 34-cycle enhanced method.

In 2001, Bruce Budowle and colleagues from the FBI Laboratory proposed several alternative strategies to boost STR profile signals without increasing PCR cycle number and experiencing the concomitant increased risk of contamination (Budowle et al. 2001). These strategies include: (1) increasing the injection time on the capillary electrophoresis (CE) instrument, (2) sample salt reduction through filtration of the PCR product to remove ions that compete with the STR amplicons when being injected into the capillary or use of a formamide with lower conductivity, and (3) reducing the PCR volume to get a more concentrated PCR product.

While these signal enhancement techniques can be helpful in some situations, none are as powerful in terms of boosting sensitivity as increasing the PCR cycle number.

Increased CE Injection

Detection sensitivity may also be improved through increasing the amount of PCR product added to the CE capillary by raising the voltage of the electrokinetic injection, lengthening

the time of injection, or both. Higher levels of PCR product passing the detector in the CE instrument lead to increased peak intensities. The standard injection for an ABI 3100 or 3130xl Genetic Analyzer is 10 s at 3 kV or 30 kV-s (see Chapter 6). The LGC Forensics DNA SenCE technique (Forster et al. 2008) involves increased CE injections (30 s at 4 kV or 120 kV-s) as well as sample desalting (see next section) while the Orchid Cellmark EnhanCE (Roeder et al. 2009) approach adds nine times the amount of purified PCR product to the sample subjected to the electrokinetic injection. ESR uses 15 s at 5 kV or 75 kV-s for their enhanced injection (Petricevic et al. 2010) while NFI utilizes 15 s at 9 kV or 135 kV-s (Westen et al. 2009).

Post-PCR Purification

As discussed in Chapter 6, the quantity of DNA injected into a CE capillary using electrokinetic injection is inversely proportional to the amount of salt present in the sample. An easy way for improving signal is to lower the salt levels in the sample. Salts such as potassium chloride and magnesium chloride are a necessary part of the buffers used with DNA polymerase to perform PCR. Thus, post-PCR purification is performed to remove salts when they are no longer needed.

This purification process requires an extra step and additional expense but works well. Several products have been used including QIAGEN's MinElute (Smith & Ballantyne 2007, Forster et al. 2008), Millipore's Montage (Roeder et al. 2009), and Edge BioSystems' Performa DTR gel filtration columns (Westen et al. 2009). PCR products can also be diluted into low-conductivity formamide to aid sample injections. With all of these salt-level reduction techniques, CE signal will improve. Internal validation is necessary to verify performance and to set stochastic thresholds (see Chapter 7).

Reduced PCR Volume

Reducing PCR volume effectively concentrates the reagents and sample being amplified. Slight improvements in PCR amplification can be seen when using a smaller volume (Leclair et al. 2003). A disadvantage is that PCR inhibitors are also concentrated with a smaller volume PCR reaction, which may lead to amplification failure or reduced efficiency.

Nested PCR

Although not widely used anymore due to the possibility of contaminating a laboratory with PCR product, nested PCR has demonstrated the ability to improve detection sensitivity with low levels of DNA template (Strom & Rechitsky 1998). Nested PCR typically involves two separate PCR reactions on the same DNA template with a transfer step in between. The first round acts as a booster for the second round, which often utilizes different primers that bind internally and create smaller PCR products than the first-round target amplicon. A major problem with this approach is that amplification tubes must be opened to introduce new reagents for the second round or to transfer a portion of the first round amplicons to the second round PCR tube. Once a laboratory is contaminated with PCR product it is very difficult to get rid of it. This, of course, is why pre- and post-PCR areas are separated physically in forensic DNA laboratories (see Chapter 4).

Altering PCR Parameters besides Cycle Number

Since 2007 commercial suppliers of STR typing kits have released next-generation kits with improved buffers and possibly an increased amount of DNA polymerase (the user does not know the amount because it is added by the manufacturer). These new buffer-polymerase master mixes have a greater ability to cope with PCR inhibitors, which improves PCR efficiency.

In addition, some kits contain miniSTR primers (see Chapter 10). Moving PCR primers closer to the repeat region can improve PCR amplification success with degraded DNA but also improves PCR yield slightly because smaller PCR products are being generated.

Mitochondrial DNA Testing

When nuclear DNA levels are very low in a sample, such as a telogen hair shaft, mitochondrial DNA (mtDNA) testing may be employed due to a higher number of mtDNA molecules per cell (see Chapter 14). The higher initial copy number improves the chances of obtaining a result in many cases. A major disadvantage with the use of mtDNA testing is the loss of specificity due to maternal relatives having identical mtDNA sequences. However, mtDNA can be very effective in aiding missing persons investigations (see Chapter 9).

CONCERNS AND CHALLENGES WITH LOW-LEVEL DNA SAMPLES

While fluorescent multiplexes have been used to obtain STR typing results from as little as a single buccal cell, a number of challenges exist when trying to obtain results from low amounts of template DNA. Attempts to generate results with low levels of DNA come up against a fundamental scientific barrier of stochastic amplification (see Chapter 4). The stochastic effects exist due to random selection of alleles when a diluted DNA template-to-primer-to-polymerase ratio exists. Even sample enrichment techniques such as whole genome amplification have not successfully overcome this problem (Schneider et al. 2004).

Stochastic Effects during PCR Amplification of Low-Level DNA

As forensic DNA analysts attempt to recover information from low amounts of DNA present in evidentiary samples, they will encounter stochastic or random sampling effects that occur in the early cycles of PCR amplification. When a limited number of DNA template target molecules exist in a sample, the PCR primers used to amplify a specific region may not consistently find and hybridize to the entire set of DNA molecules present in the PCR amplification reaction.

When enhanced detection methods are employed at least four artifacts typically arise: (1) *allele drop-in*, where additional alleles are often observed in the DNA profile from sporadic contamination, (2) *allele drop-out*, where an allele that is present in the original sample fails to amplify due to stochastic effects, (3) *increased stutter product formation*, where stutter products are often higher than the typical 5% to 10% of the nominal allele, and (4) *heterozygote peak imbalance* is often exacerbated due to stochastic PCR amplification. If one of the alleles is amplified preferentially by chance during the early rounds of PCR, then a peak

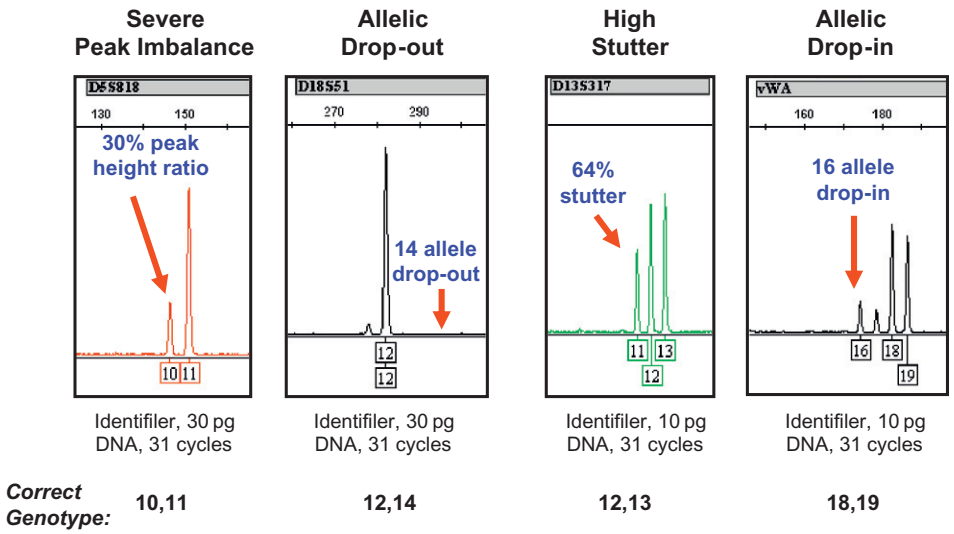


FIGURE 11.2 Stochastic effects that randomly occur when PCR amplifying low amounts of DNA using an increased number of PCR cycles. The STR typing kit, amount of DNA, and number of PCR cycles along with the correct genotype for each example are listed at the bottom. For further information on these samples, see http://www.cstl.nist.gov/biotech/strbase/LTDNA.htm.

height imbalance of greater than 60% can be observed between heterozygous alleles that should be equal or very similar in amount.

With a heterozygous locus, where two alleles are present, the unequal sampling of the alleles can result in failure to detect one or both of the alleles. Loss of a single allele is referred to as *allele drop-out* while loss of both alleles is termed *locus drop-out*. Allele drop-out can be thought of as an extreme form of heterozygote peak imbalance where one of the alleles has a peak height of zero.

Stochastic (random) variation is a fundamental physical law of the PCR amplification process when examining low amounts of DNA. Stochastic effects are manifest as a fluctuation of results between replicate analyses. In extreme stochastic sampling events, PCR amplifying the same DNA extract twice can result in different alleles being detected at a locus.

When pushing assay sensitivity through an increased number of PCR cycles, stochastic effects can become more evident. Figure 11.2 illustrates results possessing the different types of stochastic effects that may be observed when performing PCR amplification from low amounts of DNA: allele drop-out, allele drop-in, elevated stutter, and heterozygote peak imbalance.

The stochastic variation observed with the test results shown in Figure 11.2 would produce an incorrect result if no further information was available. Loss of signal, such as in the D18S51 allelic drop-out example, would make a true "12,14" heterozygote appear as a "12,12" homozygote. Likewise, the gain of signal with the high stutter or allelic drop-in could make a true single-source sample appear to be a mixture. Thus, when using enhanced

interrogation techniques, such as a higher number of PCR cycles, further testing measures are required to avoid reporting incorrect results.

Since stochastic effects cannot be avoided when testing small quantities of DNA, there are essentially two schools of thought on how to handle these types of samples: (1) stop testing or interpreting data before you go low enough to be in the stochastic realm, or (2) try to limit the impact of the stochastic variation by additional testing and careful interpretation guidelines based on validation studies (Butler & Hill 2010). Those who advocate the second approach usually enhance their method sensitivity, such as increasing the number of PCR cycles, in order to get as much out of the limited sample as possible. The "enhanced interrogation" approach typically involves replicate testing and the development of consensus profiles.

APPROACHES TO IMPROVING RELIABILITY OF RESULTS

When working with low amounts of DNA, there are three primary areas in which to potentially improve the reliability of results obtained: (1) improving DNA recovery at the collection and extraction stages so that there is more starting material, (2) improving recovery of correct profiles at the analysis stage through replicate PCR amplifications, and (3) accounting for potential allele drop-out and other stochastic effects during statistical interpretation of data acquired.

Improving DNA Recovery

DNA recovery can potentially be improved at several stages in the sampling process. Some of the cells collected on a swab during evidence sampling may not be fully released from the swab during DNA extraction. In addition, the DNA found within recovered cells may not be fully released upon DNA extraction or may bind to the tube walls during sample storage. When DNA extraction requires multiple washes or steps involving fluid transfer, a portion of the sample may be lost at each step along the way. Direct PCR (see Chapter 4) has the potential to greatly improve success rates with low amounts of starting material because no transfer steps are involved.

Researchers at the NFI found that a nylon flocked swab provided a six-fold improvement in cellular elution compared to standard cotton swabs that appear to retain sperm cells (Benschop et al. 2010). The NYC OCME showed that the addition of 1 ng carrier RNA to the Microcon filter used to concentrate the DNA aided sample recovery (Schiffner et al. 2005). While progress has been made in this area, there is probably still room for improving DNA recovery efficiencies.

Replicate Testing and Consensus Profiles

The area that has gotten the most attention for trying to improve the reliability of low-level DNA testing is replicate testing. In order to avoid or limit the possibility of getting the wrong answer when testing low amounts of DNA, replicate PCR amplifications are performed and a consensus profile developed (Taberlet et al. 1996). The standard practice is to PCR amplify either two or three aliquots of a DNA extract (Gill et al. 2000, Caragine et al. 2009).

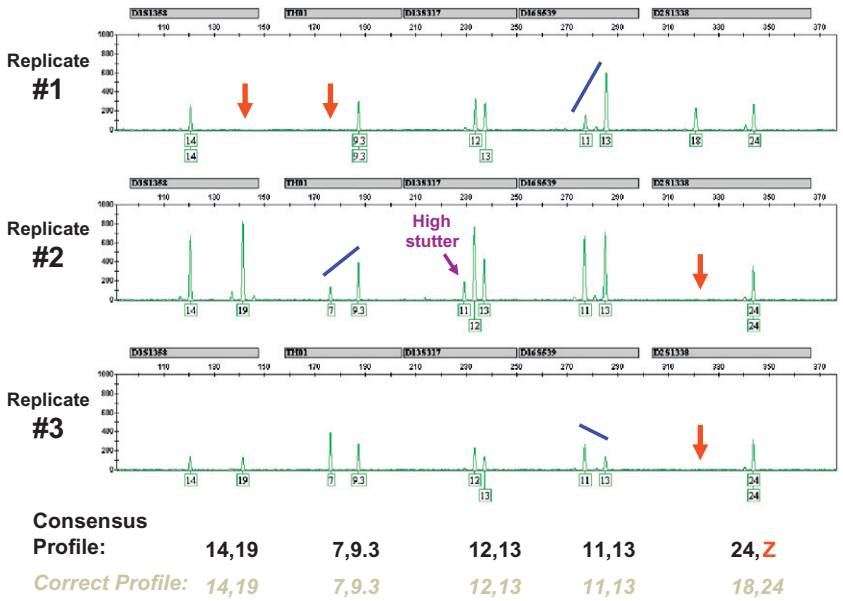


FIGURE 11.3 Three replicate PCR amplifications of a 10 pg single-source DNA template using the Identifiler kit (only green loci shown) and 31 cycles. The consensus profile, produced by recording all alleles that occurred at least two out of three times, matched the correct profile (note that the consensus wildcard "Z" for D2S1338 appropriately covers the allele 18 that has dropped out twice). The red arrows indicate positions of allele drop-out and the blue lines where severe peak height imbalance was observed.

Alleles that occur more than once (i.e., repeat) in the obtained profiles are deemed "reliable" as they have been reproduced in separate DNA tests. Based on observations during validation studies, another layer of interpretation may be applied as well before the final consensus profile is reported. For example, specific loci, such as those larger in size, may exhibit a higher rate of allelic drop-out. When reporting results from these loci, a wild card designation may be used in conjunction with a repeated single allele in order to account for potential allelic drop-out (e.g., "12,F" or "12,Z" instead of "12,12").

Three separate PCR amplifications of a 10 pg fully heterozygous DNA template using the Identifiler STR kit with 31 cycles are shown in Figure 11.3. The first locus in this figure, D3S1358, has a "14" in the first replicate, a "14,19" in the second replicate, and a "14,19" in the third replicate. The arrow indicates where the "19" allele should be in the first replicate. It has failed to amplify (i.e., allele drop-out) due to stochastic sampling effects present when only approximately two cells are available for testing. If the first replicate was the only one examined, then the D3S1358 genotype for this sample would incorrectly be designated "14,14". However, by looking for repeating alleles across the three replicates, the correct type of "14,19" can be determined as the "14" occurs in all three replicates and the "19" occurs in two out of the three.

While stochastic effects exist whenever low amounts of DNA are being examined with PCR, a replicate amplification approach with development of a consensus profile from the repeated alleles can produce reliable results (Figure 11.4). However, amplification results

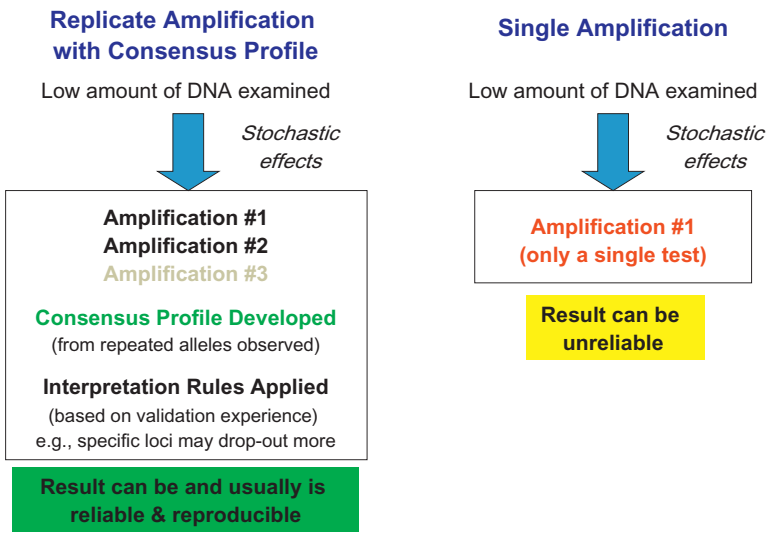


FIGURE 11.4 Comparison of approaches when examining low amounts of DNA. Replicate amplification with development of consensus profiles and application of interpretation rules based on validation experience can lead to reliable results.

from a single test can be unreliable due to allelic drop-out or allelic drop-in as noted earlier. As seen in Figure 11.3, individual results from replicate tests may vary but a combined consensus profile can generate an accurate answer when repeated alleles are recorded.

Researchers at the Netherlands Forensic Institute have studied the role of replicate number and the requested level of reproducibility through examining various combinations of six independent amplifications of known donor samples (D.N.A. Box 11.4).

NIST Data on Replicate Amplifications with Low-Level DNA

In order to evaluate reliability of results with low levels of DNA template separate from the effects of DNA degradation or PCR inhibition that might be present in forensic specimens, validation experiments were conducted at the National Institute of Standards and Technology (NIST) with pristine DNA samples (Butler & Hill 2010). Two single-source DNA samples, which were fully heterozygous at all of the tested loci, were quantified using the Quantifiler kit (Applied Biosystems, Foster City, CA) with a calibration curve created by Standard Reference Material (SRM) 2372 Component A (NIST, Gaithersburg, MD).

Following DNA quantification, dilutions were made to enable testing of 100 pg, 30 pg, and 10 pg of DNA template in each PCR reaction. A total of 10 separate PCR reactions were performed for each sample in order to study the impact and value of replicate amplifications. While 10 replicates would not be practical to perform in a casework setting with limited forensic evidence, the extra studies are valuable in a validation context to examine if more than three replicates are helpful. Both the AmpFlSTR Identifiler kit (Applied Biosystems) and PowerPlex 16 HS kit (Promega Corporation, Madison, WI) were examined using half

D.N.A. BOX 11.4

REPLICATE NUMBERS, CONSENSUS PROFILES, AND GENOTYPING RELIABILITY

In the most thorough study of low template DNA conducted to date, researchers at the Netherlands Forensic Institute (NFI) performed six separate amplifications of low-level DNA samples from known donors. They then assembled consensus profiles using between two and six of the replicates and studied the impact of allele accuracy when requiring one of four possibilities: (a) that all alleles matched between each replicate, (b) that all but one replicate matched, (c) that at least half of the replicates matched, or (d) that the allele was observed twice. Each of these samples was studied with three different methods: (a) 28-cycle PCR with standard capillary electrophoresis (CE) injection, (b) 28-cycle PCR with sample desalting and higher CE injection (9 kV), and (c) 28 + 6 cycles (i.e., for a total of 34 PCR cycles) with standard CE injection. Stochastic effects were examined and the rate of allele drop-out and drop-in measured in 414 Identifiler single-donor profiles (obtained from 23 individual samples amplified six times and analyzed with the three different methods), 90 Identifiler mixed-donor profiles (five samples amplified six fold and analyzed with three methods), and 120 MiniFiler single-donor profiles (four DNA concentrations amplified 30 times and analyzed with one setting).

A computer program was written to examine all of the combinations of the DNA profiles generated with a variable level of requested reproducibility. For example, in a sample with three replicate amplifications there are 20 different combinations possible to consider with the four different types of consensus profiles studied. In total, the NFI researchers evaluated 6578 consensus profiles for the 23 single-donor samples and 1430 consensus profiles for the five two-person mixtures. The consensus profiles obtained from DNA extracts of five single-donors were also compared to an anonymized copy of the Dutch DNA database containing 86,277 offender profiles and 40,065 forensic profiles using a "moderate" stringency search to account for potential allele drop-out (see Chapter 8). Some of the findings of this study include:

- Allele drop-in was observed in 258 instances across the 414 Identifiler single-donor profiles of which about two-thirds came from the 34-cycle PCR method and one-third from the 9 kV CE injection method. About 74% of the time (192 out of 258), the drop-in allele occurred at a stutter position (104 at N + 4 and 88 at N − 4).
- Allele drop-out was found to be largely dependent on the amount of DNA tested. Only 41 of the 414 Identifiler single-donor profiles (10%) resulted in a profile without any allele drop-out across the 16 loci amplified (19 of the 41 came from the 9 kV method and 22 from the 34-cycle method). Aliquots involving DNA input levels of 10 pg, 15 pg, 20 pg, and 40 pg were amplified eight-fold with MiniFiler. Allele drop-out decreased as DNA input levels increased such that 10 pg observed a 44% drop-out, 15 pg observed a 27% drop-out, 20 pg observed a 21% drop-out, and 40 pg observed only a 3% drop-out rate.
- The locus drop-out rate corresponded well to expected values based on

D.N.A. BOX 11.4 (*cont'd*)

allele drop-out rates. In addition, STR allele peak heights can provide some information regarding how likely it is for allele drop-out of a true heterozygote sister allele to have occurred.

- The percentage of detected alleles observed in the standard STR typing method (28 cycles) is informative to decide on which low template method (9 kV or 6 additional PCR cycles) will be most effective. When >50% of the alleles are detected in the standard method, then increased CE injection is preferred. Whereas when <10% of the alleles are detected with 28 cycles, then the increased PCR cycle number approach works best.
- Consensus method studies found that three replicates with a requirement for allele duplication work well when the standard method (28 cycles) detects >25% of alleles. For profiles with <25% of alleles detected, then a fourth replicate is helpful. Increasing the replication number to six does not yield significant gains in reliability.
- Database search studies found that wrongful homozygote designations (due to allele drop-out of a true heterozygote sister allele) can lead to a large number of adventitious matches.

Source:

Benschop, C.C.G., et al. (2010). Low template STR typing: effect of replicate number and consensus method on genotyping reliability and DNA database search results. Forensic Science International: Genetics (in press) *(doi:10.1016/j.fsigen.2010.06.00).*

reactions to conserve costs and improve sensitivity. Identifiler was tested at both 28 (standard) and 31 (enhanced) cycles. PowerPlex 16 HS was tested at 31 (standard) and 34 (enhanced) cycles.

The resulting 240 electropherograms were examined to assess stochastic effects such as allelic drop-out and allelic drop-in. These electropherograms are all available for review (as PDF files) at http://www.cstl.nist.gov/biotech/strbase/LTDNA.htm. With 16 loci for each result, a total of 3840 loci (7680 alleles) could potentially be scored. In this analysis, all peaks above 50 relative fluorescence units (RFUs) were called without an attempt to filter genotypes based on heterozygote peak height ratios.

Figure 11.5 illustrates the results for one of the tested samples with the Identifiler kit using 28 and 31 cycles. Note that use of a higher number of cycles (31 cycles) resulted in more correct genotypes as denoted by the green squares. The three extra PCR cycles improved the success rate for a correct heterozygous call from 60% (290/480 possible) to 88% (423/480 possible). The improvement from locus drop-out to correct genotype (Figure 11.5, red-to-green squares) was most dramatic at the 10 pg level where full genotype recovery improved from 4% (7/160) to 68% (108/160). With 31 cycles, allelic drop-out across the three DNA amounts tested fell from 14% (65/480) to 9% (43/480). Locus drop-out was reduced from 26% (125/480) to 2% (10/480). Thus, boosting the cycle number did improve the sensitivity and the overall success rate.

However, there were four instances of allele drop-in (Figure 11.5, black squares) when a higher number of PCR cycles were used while none existed with the lower number of cycles.

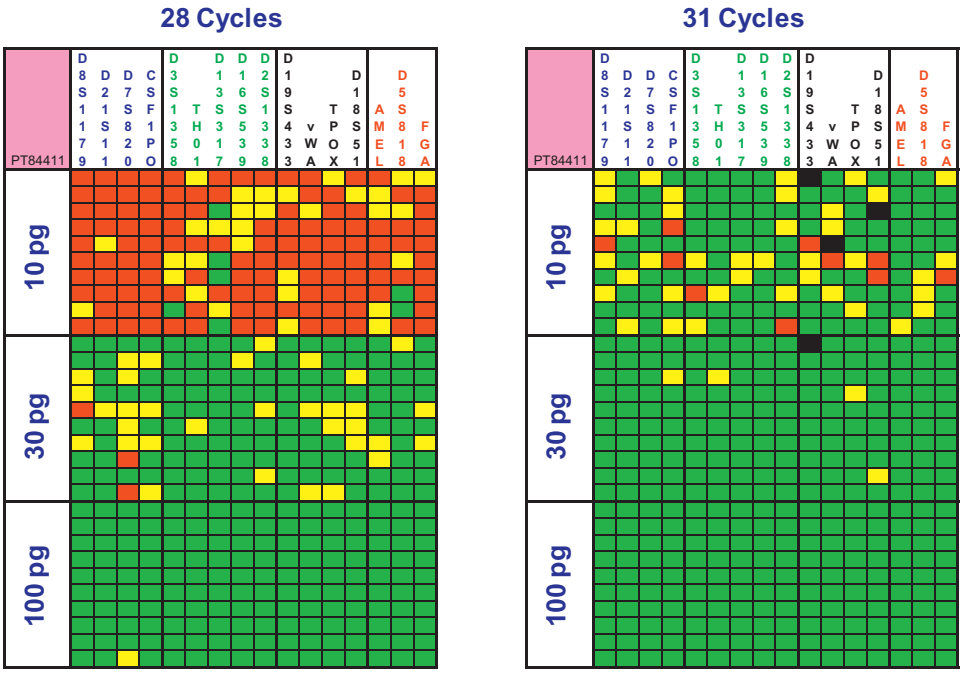


FIGURE 11.5 Sensitivity and performance summaries for Identifiler replicates with 28-cycle (standard) and 31-cycle (enhanced) data. A horizontal slice represents information from a single sample whereas a vertical slice represents the results from each of 10 replicates with the three different DNA template amounts. Green squares indicate that the full correct type was observed for that locus and PCR replicate. Yellow squares denote allele drop-out where one of the expected alleles is missing. Red squares highlight locus drop-out where both expected alleles are missing. Black squares indicate allele drop-in where an incorrect allele is observed above a 50 RFU (relative fluorescence unit) analytical threshold.

The potential occurrence of allele drop-in shows the importance of replicate amplifications and development of consensus profiles to avoid miscalls when utilizing enhanced interrogation techniques. Two of the allele drop-ins noted in Figure 11.5 are in D19S433 where the correct genotype is "13,15" yet a "13,14,15" was called with the "14" likely coming from elevated stutter of the "15" allele. The D18S51 drop-in event also appears to have arisen from high stutter. The vWA drop-in occurred at the allele "16" position when the correct type was "18,19" (see Figure 11.2).

The other sample tested with Identifiler at 31 cycles had two instances of allelic drop-in—both as a result of higher than expected stutter product formation (see http://www.cstl.nist.gov/biotech/strbase/LTDNA/ID31-summary.pdf and Figure 11.2). For the two PowerPlex 16 HS samples, one had a single observed incidence of allele drop-in at a stutter position in D3S1358 and the other sample exhibited four instances of allele drop-in at stutter positions (see http://www.cstl.nist.gov/biotech/strbase/LTDNA/PP34-summary.pdf). Since each STR typing kit will perform differently, the specific conditions and kit being examined in a forensic DNA laboratory need to be internally validated (see Chapter 7). If it is discovered that certain loci are more prone to allele and locus drop-out, then appropriate interpretation guidelines can be put in place to compensate.

In all cases with the pristine single-source DNA samples examined at NIST, replicate testing of *sequential* amplifications followed by development of consensus profiles successfully excluded any incorrect calls due to allelic drop-in. Equally important is that across *any* group of three replicates, there was never an instance of an incorrect allele being called when two of the three replicates alleles were the same. As seen in Figure 11.3, with the D2S1338 locus, duplicate single alleles were labeled with a wildcard designation to account for potential allele drop-out.

Obviously, if a limited starting amount of DNA was recovered from a sample, then there would not be enough DNA available for 10 amplifications. Moreover, if there was enough DNA available, then it would probably be used in a single high (or higher) copy number DNA analysis rather than splitting the sample.

In the world of forensic casework, DNA degradation or PCR inhibition may be a complicating factor along with the low levels of DNA template being recovered from a evidentiary item. When mixtures are observed at low DNA amounts, the individual components to the mixture will be even lower in amount and the stochastic effects become worse making it extremely challenging to recover the correct profile of the original contributors to the mixture. In these situations where the full profile cannot be recovered even with replication and consensus, statistical methods accounting for allele drop-out are an option.

Statistically Accounting for Allele Drop-out and Drop-in

In the seminal 2000 article on applying low copy number DNA techniques to forensic samples, Peter Gill, John Buckleton, and colleagues at the Forensic Science Service suggested a statistical approach that modeled the incidence of allele drop-out, laboratory contamination, and stutter (Gill et al. 2000). They envisioned that an expert system would be developed to apply their model so that the probabilities of allele drop-out and drop-in could be incorporated into the strength of evidence calculation for a crime scene profile.

While this statistical approach underpins the concepts behind replicate testing and consensus profile development (sometimes referred to as the "biological model"), the probability of drop-out is not routinely used by forensic laboratories as of early 2011. However, an approach for estimating the probability of allelic drop-out has been reported (Tvederbrink et al. 2009), and David Balding has written some free software that computes single-locus likelihood ratios allowing for drop-out of one or both alleles and drop-in of up to two alleles (Balding & Buckleton 2010). This topic will be covered more extensively in the forthcoming volume *Advanced Topics in Forensic DNA Typing: Interpretation*.

ASSORTED TOPICS

Features and Requirements of a High-Sensitivity Laboratory

Creating and maintaining a high-sensitivity laboratory that can perform LTDNA testing requires a lot of effort. Dr. Theresa Caragine from the NYC OCME DNA Laboratory spoke at a technical leaders meeting held in conjunction with the 21st International Symposium on Human Identification in October 2010. From her experience running a high-sensitivity

laboratory that examines evidence in hundreds of low-level DNA cases each year, she offers five tips to preventing exogenous DNA contamination in forensic casework. These same tips are valuable for laboratories performing mitochondrial DNA testing as well.

First, utilize dedicated and protected workspaces such that each step in the DNA process has a different hood space. Always process the known reference samples separate from the unknowns. Conduct the evidence examination, pre-PCR, and post-PCR testing in separate rooms.

Second, wear proper personal protective equipment (PPE). This includes a laboratory coat, head covering, a mask or face shield, booties, eye protection, and double gloves. The primary reason for such extensive PPE is to protect the DNA samples from the staff working on them. The double gloves prevent skin contact. Only the outer gloves are changed in the laboratory between each item of evidence and after touching refrigerator handles or other items that could potentially cross-transfer any cellular material. Lab coats are single-use within the evidence exam area. Lab coats remain in the pre-PCR or post-PCR areas and each one is used by a single analyst for typically only a few days.

Third, perform regular cleaning of *everything*. Careful cleaning of the entire lab is conducted weekly and documented with a checklist. All surfaces are cleaned with 10% bleach, which is typically made daily to keep it fresh. Surfaces are then scrubbed with water and 70% ethanol after the bleach treatment. Hoods, pipets, and instruments are cleaned between each assay and piece of evidence. Aerosol-resistant pipet tips are used and changed with each new sample to prevent cross-contamination during liquid transfers. Scissors, tube openers, tweezers, and pens are all dedicated to the room where they are used (i.e., they do not leave the room) to prevent inadvertent transfer of any DNA material. A UV light is applied in each hood for 15 minutes before and after use. Tissue culture techniques are employed meaning that the pipet and the hands of the scientist never cross over open tubes or reagents when setting up assays. Only one sample tube is opened at a time to prevent cross-contamination. Sample tubes are also opened with a cap opener or a clean tissue.

Increased sensitivity with LTDNA or mtDNA techniques means that minute levels of contaminant DNA in labware or reagents may be detected from the manufacturer that would not be seen with conventional STR typing. Thus, all labware and reagents are UV-irradiated prior to use in PCR to try to inhibit the replication of exogenous DNA. All plasticware, instrumentation, extraction reagents, and even the water used is irradiated (Tamariz et al. 2006). If the item cannot be irradiated, then it is bleached.

Fourth, perform quality control testing on all reagents prior to use on evidentiary items. NYC OCME utilizes a multi-copy Alu qPCR assay for high-sensitivity DNA detection (see Chapter 3). Each batch of reagents must be cleared for use by showing no result with the qPCR assay as well as no detectable alleles with the enhanced detection LTDNA STR analysis. To ensure appropriate sensitivity for the STR typing kits, 100 pg, 25 pg, and 6 pg of a postive control DNA template are tested to determine if each amount yields an expected number of alleles as previously seen in their validation studies.

Fifth, evaluate controls with every batch of samples. The positive control tests whether the assay is robust and working properly. The negative control highlights any contamination that might be present in the PCR reagents. Negative controls are treated in as sensitive a manner as the tested samples and are amplified in triplicate as are the LTDNA samples in order to develop a consensus profile. Some stray alleles are occasionally detected in

negative controls. During their validation studies, NYC OCME found that 89% of the time the negative controls were clean. However, most of the detected alleles in the remaining 11% of the samples were typically random and not due to gross contamination. If a labeled allele was found to repeat in two of three replications for the negative control, then any sample having that labeled allele at that locus is declared inconclusive for that locus. If there are more than two repeating alleles over the three replicates for a negative control, then the batch fails and testing would have to be repeated if sufficient sample remains.

Another important measure to detect potential contamination when using enhanced detection methods is an elimination database for comparison purposes. All lab staff, all cleaning and facilities staff, and all members of the local crime scene unit are DNA tested. Their DNA profiles are placed on the elimination database and compared to evidentiary profiles prior to completing a forensic case report. These profiles are never uploaded to the DNA database. Rather anyone who may have come in contact with the laboratory space or the original biological sample is eliminated first as a possible contaminant.

Minor Components in Mixtures May Be Low-Level DNA

Many DNA analysts may think that low-level DNA analysis does not apply to them because they are running 28-cycle PCR and not examining DNA down at a level of 100 pg or less. However, PCR amplification involving 1 ng total DNA with a two-person mixture and a 9:1 major-to-minor component ratio leaves the minor contributor in the low template range of approximately 100 pg or 15 cells (Table 11.3). Thus, minor contributor alleles in this situation could be experiencing stochastic sampling (allele drop-out, etc.) as well as allele masking by the taller major contributor alleles. This fact is important to keep in mind when working with DNA mixtures.

Validation Experiments

The low-level DNA characteristics of allele drop-out, allele drop-in, higher stutter, and greater heterozygote peak height ratio imbalance mean that internal validation experiments need to be performed in order to understand the limits of the specific technique being applied. Table 11.4 summarizes some of the published validation work performed by LCN DNA laboratories around the world. Note that each lab has selected different PCR conditions or CE injection parameters as optimal.

When performing validation studies, it is best to utilize samples that are heterozygous at all tested loci if possible so that allele drop-out can be monitored. The height of the surviving sister allele from the heterozygote is useful in determining where (or whether) a stochastic threshold can be set. The ESR validation paper (Petricevic et al. 2010) may serve as a useful model for plotting data obtained from validation experiments.

Examination of multiple samples is equally important as DNA quantitation values on the tested samples (that are then diluted to low levels for testing) may not be exact. Observing PCR performance in a range of samples, which may have slightly different true amounts of DNA, is valuable in setting interpretation guidelines based on the validation data obtained.

TABLE 11.3 Correlation of DNA Amount with the Approximate Number of Diploid Cells Used in Single-Source and Mixture Samples. Calculations for 1 ng Being Equivalent to 152 Diploid Cells May Be Found in D.N.A. Box 3.1. Shaded Boxes Contain Less than 20 Cells and Therefore May Exhibit Stochastic Effects (Walsh et al. 1992). Adapted from a 2003 Presentation by Robin Cotton (Then of Orchid Cellmark) at the American Academy of Forensic Sciences Low Copy Number DNA Workshop.

Single-Source *Amount of DNA*	*Total Cells in Sample*	*Approximate Number of Copies of Each Allele if Heterozygous*
1 ng	152	152
0.5 ng	76	76
0.25 ng	38	38
0.125 ng	19	19
0.0625 ng	10	10
1:1 Mixture *Amount of DNA*	*Total Cells in Sample*	*Approximate Number of Cells from Each Component*
1 ng	152	76
0.5 ng	76	38
0.25 ng	38	19
0.125 ng	19	10
0.0625 ng	10	5
1:3 Mixture *Amount of DNA*	*Approximate Number of Cells from* **Major Component**	*Approximate Number of Cells from* **Minor Component**
1 ng	114	38
0.5 ng	57	19
0.25 ng	28	10
0.125 ng	14	5
0.0625 ng	7	2
1:9 Mixture *Amount of DNA*	*Approximate Number of Cells from* **Major Component**	*Approximate Number of Cells from* **Minor Component**
1 ng	137	15
0.5 ng	68	8
0.25 ng	34	4
0.125 ng	17	2
0.0625 ng	9	1

TABLE 11.4 Examples of Some Validation Efforts with Low-Level DNA.

Laboratory	STR kit	Summary of Approach & Results	Reference
Forensic Science Service (UK)	SGM Plus	Gill: 21 of 31 negative controls showed minor levels of STR contamination; pair-wise comparisons performed on replicate amplifications found that 4 of 1225 (0.3%) showed a spurious allele being duplicated. Whitaker: Of 577 heterozygote loci tested, 37 showed locus drop-out and 58 showed allelic drop-out.	Gill et al. (2000), Whitaker et al. (2001)
Armed Forces DNA Identification Laboratory (U.S.)	PowerPlex 16 & Yfiler	Added twice the amount of Taq polymerase and added six PCR cycles (up to 36 total) to improve human remains identification; used triplicate amplifications with allele duplication required for reporting; ~30% allele drop-in observed; obtained results from 90-year-old specimen.	Irwin et al. (2007), Sturk et al. (2009)
LGC Forensics (UK)	SGM Plus	Forster: Compares use of 34-cycle PCR run in duplicate with DNA SenCE (Sensitive Capillary Electrophoresis), which is 28-cycle PCR with MinElute post-PCR clean-up and enhanced injection (30 s @ 4 kV compared to standard 10 s @ 3 kV), on 36 mock evidential items; observed a 0.7% drop-in rate per locus. Gross: Reviewed 1398 control samples processed alongside casework samples run over a five-month period; 12% (170/1398) had one or more drop-in peaks with a total of 255 extra alleles in 7040 loci for a drop-in probability of 3.6% per locus.	Forster et al. (2008), Gross et al. (2009)
Orchid Cellmark (UK)	SGM Plus & Identifiler	Examined 42 samples with various combinations of 28, 30, and 34 PCR cycles and Montage post-PCR clean-up with either 1 µL of PCR product added to 9 µL Hi-Di formamide (Phase 1 enhancement) or 9 µL of purified PCR product with 11 µL of Hi-Di formamide (Phase 2).	Roeder et al. (2009)
New York City Office of Chief Medical Examiner	Identifiler	Used 0.01% SDS to limit PCR inhibition and carrier RNA to improve recovery during DNA extraction; PCR reaction volume of 13 µL with 31 cycles (instead of 28 cycles); increased CE injection time and voltage (30 s @ 6 kV); samples were amplified in triplicate and all alleles above 75 RFU were scored with consensus profiles being developed when alleles were labeled in at least two of the three replicates; generated accurate allele calls on 319 forensic casework samples; developed a flow chart to aid in LT-DNA interpretation.	Caragine et al. (2009)
Institute of Environmental Science and Research (ESR) (New Zealand)	SGM Plus	Compared 1 ng DNA template results run at 28 cycles with 12.5 pg and 25 pg (amplified in duplicate) run at 34 cycles; normal CE injection (10 s @ 3 kV) compared with enhanced injection (15 s @ 5 kV); examined 20 samples that were fully heterozygous at the 10 SGM Plus STR loci; 28-cycle data was composed of 422 heterozygous loci; 34 cycle data contained 1688 heterozygous loci; average drop-in rate was 1.3% per locus; 99% of stutter products were less than 25% of the associated allele peak height; stochastic threshold set at 3000 RFU for normal CE injection and 7000 RFU for enhanced injection; encouraged use of models that place less emphasis on peak height differences with mixture interpretation.	Petricevic et al. (2010)
Netherlands Forensic Institute (NFI)	Identifiler	See D.N.A. Box 11.4.	Benschop et al. (2010)

Sampling Limitations with Sperm Cells

New sampling techniques such as laser capture microdissection (see Chapter 2) enable collection of specific cells. Keep in mind that individual sperm cells contain only half of the genomic material from the donor. Therefore, multiple sperm cells will need to be collected in order to represent the complete DNA profile (D.N.A. Box 11.5).

Whole Genome Amplification

A DNA enrichment technology known as whole genome amplification (WGA) has been explored as a possible method for recovery of limited quantities of DNA from evidentiary samples (Bergen et al. 2005, Hanson & Ballantyne 2005, Ballantyne et al. 2007). WGA involves a different DNA polymerase (phi29) than the TaqGold enzyme commonly used in forensic DNA analysis and amplifies the entire genome using random hexamers as priming points. The WGA enzymes work by multiple displacement amplification (MDA), which is sometimes referred to as rolling circle amplification. MDA is isothermal with an incubation temperature of 30°C and requires no heating and cooling like PCR.

QIAGEN (Valencia, CA) and Sigma-Aldrich (St. Louis, MO) both offer phi29 DNA polymerase cocktails for performing WGA. The kit sold by QIAGEN is called REPLI-g while Sigma-Aldrich's kit is GenomePlex. Yields of 4 μg to 7 μg of amplified genomic DNA are possible from as little as 1 ng of starting material. The phi29 enzyme has a high processivity and can amplify fragments of up to 100 kb because it displaces downstream product strands enabling multiple concurrent and overlapping rounds of amplification. In addition, phi29 has a higher replication fidelity compared to Taq polymerase due to 3′–5′ proofreading activity.

While all of these characteristics make WGA seem like a possible solution to the forensic problem of limited DNA starting material, studies have found that stochastic effects at low levels of DNA template prevent WGA from working reliably (Schneider et al. 2004). Allele drop-outs from STR loci were observed at 50 pg and 5 pg levels of starting material (Schneider et al. 2004) just as are seen with current LT-DNA testing. Work with "molecular crowding" materials such as polyethylene glycol, where the amount of DNA is enriched in localized areas of a sample, has shown improved success with STR typing from low amounts of DNA (Ballantyne et al. 2006).

It is possible that WGA may play a limited role in enriching samples for archiving purposes that are in the low ng range (Lasken & Egholm 2003), but it will probably not be the end-all solution to LT-DNA samples in the low picogram range. Thus, it appears, as with regular PCR techniques, stochastic selection of alleles present in low-level DNA samples limits the effectiveness of WGA to several hundred picograms in order to recover a full profile with a single amplification attempt.

Caution in Relying on DNA Quantitation Values

Is it possible to ascertain that sufficient DNA material exists to obtain reliable results where the DNA profile obtained is expected to accurately reflect the source of the biological sample? There are two primary points in the DNA testing process where potential DNA reliability may be assessed: (1) at the DNA quantitation stage prior to performing PCR

D.N.A. BOX 11.5

PROBABILITY OF ACHIEVING FULL ALLELIC REPRESENTATION FOR LOW NUMBERS OF HAPLOID CELLS

Sexual assault evidence is composed of sperm cells, which contain only half of the full complement of genomic DNA from the donor male because of meiosis. Thus, to obtain a representative *diploid* DNA profile containing both possible alleles from the original sample source, multiple sperm will need to be analyzed. Techniques such as laser capture microdissection enable collection of individual sperm cells (Elliott et al. 2003, see Chapter 2). When working with limited amounts of evidentiary material, the question becomes how many individual sperm cells are needed to obtain a fully representative DNA profile when multiple loci are being examined.

In a 2007 paper published in the journal *Science & Justice*, David Lucy and colleagues found that 19 haploid cells would be theoretically required to obtain a full 15-locus DNA profile with a 99.99% probability (Lucy et al. 2007). Since each sperm cell contains just over 3 pg of DNA (see D.N.A. Box 3.1), approximately 60 pg of sperm DNA extract would therefore be required to obtain a full profile containing both alleles at all 15 tested loci.

The number of haploid cells required to obtain complete representation of a full profile at a specified number of heterozygous loci with a given probability (ranging from 90% to 99.99%) is provided in the table to the right (Lucy et al. 2007).

Number of Loci	0.90	0.95	0.99	0.999	0.9999
1	5	6	8	11	15
2	6	7	9	12	16
3	6	7	10	13	16
6	7	8	11	14	17
10	8	9	11	15	18
15	9	10	12	15	19

This information is theoretical in nature and may not necessarily reflect the probabilities of what is observed after DNA extraction, PCR amplification, and capillary electrophoresis injection. The efficiency of these processes will govern the actual numbers of cells required. Research scientists from the Forensic Science Service found that several hundred sperm cells can be required in order to have a high probability of obtaining a complete STR profile (Elliott et al. 2003). Poor extraction efficiency and less than 100% amplification efficiency due to DNA degradation or PCR inhibition play a role in this discrepancy between the theoretical predictions and empirical observations.

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amplification of the STR markers of interest or (2) by examining the peak heights—and peak height ratios in heterozygous loci—in the STR profile obtained.

An empirically determined threshold (usually termed a "stochastic threshold") may be used at either the DNA quantitation or data interpretation stage to assess samples in the potential danger zone of unreliable results. For example, if the total amount of measured DNA is below 150 pg, a laboratory may decide not to proceed with PCR amplification assuming that allelic drop-out due to stochastic effects is a very real possibility. Alternatively, a laboratory may proceed with testing a low-level DNA sample and then evaluate the peak height signals and peak ratios at heterozygous loci. When peak height ratios for heterozygous loci in single-source samples dip below 60%, there is an indication that stochastic effects are significant which would make it challenging to reliably pair alleles into major and minor genotypes with mixtures. This topic will be covered more extensively in the forthcoming volume *Advanced Topics in Forensic DNA Typing: Interpretation*.

Since the advent of quantitative PCR (qPCR) assays, DNA quantitation tests have become more sensitive—enabling quantities as small as a few genomic copies to be detected (see Chapter 3). However, it is important to keep in mind that qPCR is also subject to stochastic variation especially on the low end of DNA quantity measurement. Thus, numbers in the low picogram range may not be reliable and results with little or no "detectable" DNA may still amplify with STR kits (Cupples et al. 2009; see also D.N.A. Box 3.3).

In an early paper discussing stochastic effects and the limitations of PCR assays, Walsh et al. (1992) proposed avoiding stochastic effects by adjusting the number of PCR cycles in an assay so that the sensitivity limit is around 20 or more copies of target DNA. In other words, their goal was to enable a full DNA profile to be reliably obtained with approximately 125 pg of DNA. Below roughly that amount, allele and locus drop-out would be expected and partial DNA profiles would result (Walsh et al. 1992). Obtaining a partial DNA profile is an indication that a low-level DNA amplification has occurred.

Depending on the STR typing kit primer and DNA polymerase concentrations and the fluorescent dye sensitivities, the number of PCR cycles is typically set by manufacturers in the range of 28 cycles to 32 cycles. However, as noted previously, STR kits certainly work beyond manufacturer recommended cycle numbers. If laboratories choose to increase cycle numbers beyond what is recommended by manufacturers, validation studies are needed to help set appropriate interpretation guidelines.

Thresholds Are Difficult to Set with Enhanced Detection Methods

Stochastic thresholds, such as 150 RFU or even 500 RFU, may not apply for enhanced detection methods that include increasing the number of PCR cycles. Instead, independent replicate amplifications and concensus profile development are necessary to compensate for allele drop-out and drop-in. Computer software that involves probabilistic modeling to data, such as LoComatioN (Gill et al. 2007), will be important to future advances with low-level DNA analysis.

STRBase Website on Low Template DNA

A low template DNA section of the NIST STRBase website was launched in October 2009 following the International Symposium on Human Identification LCN Panel. This website,

which is available at http://www.cstl.nist.gov/biotech/strbase/LTDNA.htm, contains three primary sections: (1) presentations, (2) NIST data from sensitivity studies, and (3) a listing of literature on the topic.

With the LTDNA website launch, four presentations given by John Butler (NIST), Becky Hill (NIST), Theresa Caragine (NYC), and Charlotte Word (consultant) at the ISHI 2009 meeting were included. As previously mentioned, data from low-level DNA experiments using Identifiler and PowerPlex 16 HS kits with 100 pg, 30 pg, and 10 pg and different numbers of PCR cycles are available to be viewed as PDF files. The two DNA samples examined in these studies were fully heterozygous at all STR loci providing an opportunity to monitor peak imbalance and allelic drop-out under different conditions. Laboratories are encouraged to submit their validation data for inclusion on this community resource as suggested previously (Buckleton 2009, Buckleton & Gill 2011).

A literature listing of pertinent articles to help explain the issues involved with low-template DNA testing is provided on the STRBase LTDNA website and will continue to be updated over time. The articles are listed according to four categories in order to reflect their relative reliability in scientific terms: (1) peer-reviewed literature, (2) reports, (3) reviews, and (4) non-peer-reviewed literature. In science, as in other fields, not all information is equally authoritative or helpful. Thus, the literature on low-template DNA analysis is broken into several categories on the STRBase website in order to reflect the variation in scrutiny and support.

Summary and Final Thoughts

Every lab faces samples with low amounts of DNA. The choice becomes whether or not to attempt an "enhanced interrogation technique" such as increasing the cycle number, desalting samples, or injecting more sample onto the CE. If so, are there sufficient validation studies to underpin this work and to develop appropriate interpretation guidelines? Generating STR data with an increased number of PCR cycles and invoking an LCN philosophy can provide a useful lead in many instances for an investigation but it is unlikely to provide definitive probative evidence of a crime in every instance.

At what point does a laboratory draw a line and not attempt to analyze DNA data below this line (i.e., to declare a result inconclusive)? Is this line based on a certain amount of input DNA? If so, what validation data underpins this quantitation threshold? Does the laboratory have a predetermined stochastic threshold used during data interpretation? Again, on what validation data is this value based?

If a laboratory chooses to perform low-level DNA analysis with enhanced detection methods, interpretation guidelines need to reflect the potential artifacts of this high sensitivity environment. Laboratories should not report out a single result with LCN STR analysis (see Figure 11.4). Allele duplication with multiple replicates and formation of a consensus profile is currently the recommended best overall approach. However, this is not a license to report any and all replicated data. Caution is needed when interpreting data and drawing conclusions from low-level data due to stochastic effects that can and will exist in the results obtained.

When working with low amounts of DNA template, success rates are often poor. Dedicated "clean" facilities and extreme care are required to avoid or at least limit contamination. Often

mixtures result from touch DNA or other low-level DNA analyses that may not yield meaningful results. Finally, in the end, results may not be probative if the sample could have been deposited innocently at the crime scene before the crime even occurred. Thus, low-level DNA recovered from a crime scene may not be relevant to the crime under investigation.

Nevertheless, research and validation experiments have demonstrated successful low-template DNA analyses that track appropriately back to the sample donor (e.g., D.N.A. Box 11.4). Thus, low-level DNA results can be reliable. When appropriate controls are in place and replicated results are repeatable, DNA interpretation should proceed—but cautiously.

The next-generation STR kits, such as the PowerPlex 16 HS, PowerPlex ESI 17 and ESX 17 Systems and AmpFlSTR Identifiler Plus and NGM kits, with their greater sensitivity and ability to overcome PCR inhibition, have the potential to make the current qPCR DNA quantitation kits obsolete as an appropriate gatekeeper to whether or not to continue with a low-level, compromised DNA sample. *With greater power to obtain results comes greater responsibility to report reliable results.* Careful validation studies and development of appropriate interpretation guidelines will continue to be essential as laboratories move forward with caution and care in analysis of low amounts of DNA.

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